scientific reports

Check for updates

OPEN

Seasonal solar irradiance forecasting using artificial intelligence techniques with uncertainty analysis

V. Gayathry[1], Deepa Kaliyaperumal[1]✉ & Surender Reddy Salkuti[2]

Renewable integration in utility grid is crucial in the current energy scenario. Optimized utilization of renewable energy can minimize the energy consumption from the grid. This demands accurate forecasting of renewable contribution and planning. Most of the researches aim to find a suitable forecasting model in terms of accuracy and error metrics. However, the uncertainty and variability in these forecasts are also significant. This work combines point forecast with interval forecast to provide comprehensive information about the forecast uncertainty. In this work, solar irradiance forecasting is carried out using artificial intelligence (AI) techniques. Forecasting is done using seasonal auto-regressive moving average with exogenous factors (SARIMAX), support vector regression (SVR), long short term memory (LSTM) techniques and performance is evaluated. SVR model exhibited the best performance with R^2 values of 0.97 and 0.96 for winter and summer respectively and 0.85 for monsoon and post-monsoon seasons. This is followed by forecast error distribution studies and uncertainty analysis. For this, SVR forecast error data is fitted using laplace distribution. Uncertainty study is carried out using confidence intervals and coverage rates. Excellent coverage rates are obtained for various confidence levels for all seasons, indicating the appropriate fitting of error distribution. For the narrow 85% confidence band, coverage rates of 89%, 95%, 90%, and 88% are obtained for winter, summer, monsoon and post-monsoon respectively. The work emphasizes the need for error-distribution studies, modeling of forecast errors and their application in providing reliable forecast intervals with the perspective of enhancing system reliability.

Keywords Solar forecasting, Uncertainty analysis

Energy crisis is a serious concern all over the globe in the recent years. Global warming and climate crisis add to the problem. Energy needs are increasing day by day. Traditional fossil-fuel based energy generation has its own setbacks. The major contributor for global warming is fossil fuel combustion. Replacing fossil fuel energy with renewable energy resources has several advantages. Renewable resources like solar and wind are easily available, and also pollution free at source. The economic benefits of renewable energy resources are also large. Renewable energy integration in the power and energy market is the primary solution to address the energy and climate crisis[1]. As against the conventional, centralized technology, distributed generation involves energy generation at or near user ends, mainly multiple renewable resources.

The concept of microgrid is gaining significance globally. Microgrids are low voltage grids, comprising of distributed energy sources which can operate both in grid-connected and disconnected mode. It consists of energy sources, storage units, and loads with a well-defined electrical boundary. Microgrids can operate along with the utility grid or can operate independently. These greatly support the integration of renewables to the power grid. However, intermittency in the renewable generation and highly varying load patterns pose challenges in the microgrid operation. Renewable resources like solar and wind are highly weather dependent and load patterns are human behavior dependent. A reliable management system for matching the demand and supply in the microgrid is, therefore, crucial.

Optimizing the power generation in a microgrid, including conventional and renewable resources, to supply adequate power is the need of the hour. Efficient forecasting and planning can help in reducing risks in the

[1]Department of EEE, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Bengaluru, India. [2]Department of Railroad and Electrical Engineering, Woosong University, Daejeon, Republic of Korea. ✉email: k_deepa@blr.amrita.edu

Content courtesy of Springer Nature, terms of use apply. Rights reserved

operation, thereby improving the network stability, system management, and operation. Forecasting the generation and demand is the primary step towards an efficient energy management schedule in the microgrid and is an area of research[2–4].

In this work, solar energy generation in a conceptual microgrid is forecasted using AI techniques. This is an indirect method, from the microgrid planning perspective, in which, solar irradiance is forecasted and solar power output is calculated from the predicted irradiance. Following the forecast, distribution of the forecast errors is studied and the parametric distribution that fits the error data is obtained. This helps in quantifying the uncertainty associated with the forecast.

Related work

In general, forecasting methods are of different types, namely physical, statistical and hybrid methods. Physical methods make use of climate parameters, location/site information and other parameters and describe them in physical equations, to forecast the target. Statistical methods make use of historical data and extract information from this data. These include time series auto-regressive methods, regression methods, smoothing techniques etc. Recent advances in AI techniques, involving machine learning (ML) and deep learning (DL) methods are included under statistical methods[5]. In addition, hybrid models which are combinations of one or more models have also evolved.

Solar forecasting can be direct forecasting or indirect forecasting. Direct forecasting involves forecasting the solar power output from the generation system. Indirect forecasting involves forecasting the irradiance and then calculating power from the predicted irradiance. A partial functional linear regression model is developed and validated in[6]. Forecasting using auto-regressive moving average (ARMA) models is discussed in[7]. AI techniques are widely used for supply and load forecasting in power sector in recent years[8–26]. The work in[12] discusses the forecasting of photovoltaic (PV) power using support vector machine (SVM) and linear regression. More the parameters in the learning, lesser will be the prediction error. The work in[13] employs random forest (RF) and SVM for PV output power prediction, and, compares the performances. Even though both the models capture the trends in the data, RF performs better than SVM. In terms of performance error metrics, RF performs approximately 40% better than SVM. In[14], three methods, artificial neural networks (ANN), deep neural networks (DNN) and LSTM are suggested for forecasting solar power output. When the ANN model failed to provide satisfactory results, LSTM model provided better results and is also better than conventional statistical methods compared in the work. Another work employing LSTM is discussed in[15]. The work uses a synthetic weather forecast, along with historical features for prediction. The superiority of LSTM compared to generalized regression neural network (GRNN), recurrent neural networks (RNN), and extreme learning machine (ELM) is also discussed. In[16] also, LSTM model for multiple forecast horizons are developed and compared against DNN, multi-layer perceptron (MLP), auto-regressive integrated moving average (ARIMA), SVR algorithms. Convolutional neural networks (CNN) and gated recurrent units (GRU) are also used in solar forecasting works. The work in[17] compares the DL models LSTM, BiLSTM, attention based LSTM, GRU and CNN models in forecasting irradiance data at multiple locations, for a period of 36 years.

An approach employing CNN and LSTM is carried out in[18]. CNN model discovers the features in the previous power data. LSTM model considers the power on the same date as input and forecasts the power of next time instant. Further, the forecasted values are reconstructed based on LSTM and CNN results. An ensemble method combining time series and ML techniques is detailed in[19]. ARIMA, vector autoregression (VAR), and LSTM are the models used. Simple average, MLP based and CNN based ensemble methods are employed. Feature selection techniques are carried out and CNN based ensemble method with feature selection is found to have higher prediction power.

New studies focus on hybrid models combining decomposition algorithms and AI techniques. The work in[20] employs a combination of decomposition algorithms, ML and DL models for multistage power prediction. Iterative filtering, variational modal decomposition (VMD) and wavelet packet decomposition (WPD) are the different decomposition algorithms employed. Results from the decomposition algorithms are grouped into sets and assigned to ML and DL models including SVM, ELM, gaussian process regression (GPR), ANN, LSTM, BiLSTM, GRU and CNN. The proposed work achieves very low normalized root mean square error (NRMSE) values for different forecast horizons. A robust local mean decomposition (RLMD) combined with BiLSTM is compared against benchmark models in[21]. The proposed model outperforms the benchmark models in terms of root mean square error (RMSE), R^2, and mean absolute error (MAE) metrics. Another work[22] employs empirical mode decomposition (EMD), ensemble EMD (EEMD), complete ensemble EMD with adaptive noise (CEEMDAN), VMD, wavelet transform (WT) and WPD as the decomposition algorithms. FeedForward NN, LSTM, GRU, and BiLSTM are the DL models used. Among these, WPD based model achieved the best performance with minimum RMSE and MAE values.

An ultra-short term forecasting using an improved ant-colony optimized (ACO) SVM model is discussed in[23]. The impact of proper data pre-processing on the performance metrics is also analysed. The hybrid model showed very good prediction accuracy with R^2 value up to 0.997. A similar work is detailed in[24], where in a genetic algorithm (GA) based SVM (GASVM) model is employed for short term forecasting of solar power. The work in[25] proposes another algorithm using an improved chicken swarm optimizer and ELM. The optimizer optimizes the weights of the model and improves the predictive power, with an average mean absolute percentage error (MAPE) and RMSE of 5.54% and 3.08%, respectively. A Bayesian-optimized BiLSTM/LSTM model is used for hourly solar forecasting[26,27]. The hyperparameters for LSTM models are obtained by Bayesian optimization. In[26], the work considers both univariate and multivariate architectures, considering various extraneous variables affecting the target. The proposed model offered good prediction with an R^2 value of 0.99 for both univariate and multivariate models. Another work in[28] compares two models, adaptive neuro-fuzzy inference systems (ANFIS)

Content courtesy of Springer Nature, terms of use apply. Rights reserved

and particle swarm optimization based ANN (PSOANN) in solar power forecasting. The ANFIS model showed better prediction with an accuracy of 99.85% whereas PSOANN depicted an accuracy of 98.9%. The work in[29] presents a forecasting method using sky images. Features extracted from the sky images in the preliminary stage are fed to the ML regression models, k-nearest neighbor (kNN) and RF. The proposed methods showed good prediction capability and can perform forecasting up to four hours ahead.

Most of the research works focus on finding a model that best forecasts the target based on the accuracy metrics. However, from a system point of view, understanding and quantifying the uncertainty involved in the prediction is required. Point forecasting techniques predict one single future value. However, owing to the high variability in solar irradiance, predicting a range or band of possible values will be more useful than predicting a single value. Point forecasting techniques do not provide information about the prediction uncertainty. Interval forecasting can capture the variability and therefore, the uncertainty element in the irradiance data, leading to better decision-making. This provides information not only about the possible output values, but also the possible range of variation of these values. These forecasting techniques fall into two categories, parametric and non-parametric. In parametric method, the random variable follows a known probability density distribution e.g., normal, laplace, gamma distribution. However, for non-parametric distribution, any assumption of probability distribution is not made, which is more realistic. Prediction intervals (PI) and confidence intervals (CI) are employed to quantify forecast uncertainty. Literatures[30–36] have conducted uncertainty studies on different energy forecast models. Forecasting error in a wind forecasting model is studied in[30] and fitted using different statistical distributions. Finally, the Cauchy distribution is found as the best fit. CIs are used to quantify the forecast uncertainty here. CIs for 90% and 95% are constructed and uncertainty in the prediction is studied. The authors explain the need for accurate modeling of error distribution for efficient operation and operating reserve allotment in the system. However, in this work, absolute values of the errors are used for modeling. In[31], the wind power forecast uncertainty is studied using normal, beta, hyperbolic and weibull distributions. In[32], solar irradiance forecast errors are found to follow a t-distribution. For an assumed 68% confidence interval, detailed uncertainty analysis is carried out for real time data collected from a location in Stellenbosch, South Africa. The work in[33] forecasts the solar power using a hybrid forecast model followed by a non-parametric kernel density estimation (NKDE) technique for modeling the error distribution. The PIs for different confidence levels are calculated and the results are validated on a 15kW grid connected PV system. Similarly, NKDE is carried out in[34,35] applied to wind forecasting. In[36], Monte Carlo simulations are used to calculate the uncertainty in wind speed forecasting.

It is essential that along with the point forecasts, energy forecasting models should also provide an assessment of the uncertainty associated with these forecasts. From the literature study conducted it is seen that interval forecasts and forecast uncertainty analysis are not given enough attention and focus as compared to point forecasts. Also, a considerable amount of work is in progress related to the uncertainty analysis of wind forecasting, however, this is not so in the case of solar power forecasting. Also, many of the above works, use absolute errors for modeling, which need to be transformed accordingly while understanding the confidence bands. In this regard, the proposed work tries to bridge this gap by conducting a point forecasting study of solar irradiance using well-proven models, followed by interval forecasting for quantifying the forecast uncertainty. Considering the importance of solar energy in today's power sector including microgrid and smart grids, there is a need for study and detailed research on solar prediction error modeling and uncertainty analysis. The current work tries to address this concern and hence, is more focused towards solar forecasting uncertainty analysis.

This work focus on the below objectives:

- Seasonal forecasting of hourly solar irradiance using AI models, SARIMAX, SVR and LSTM. A specific location in Bangalore, Karnataka, India is selected.
- Modeling of seasonal forecast errors using a parametric laplace distribution.
- Quantification of forecast uncertainty using evaluated confidence levels and coverage rates.

In the following pages, the first section explains the materials and methods, which includes the dataset and the algorithms used. This is followed by the implementation details and model building. The next section explains the results and discussions. Further, uncertainty analysis is carried out. Finally, conclusion and discussion regarding the future scope of the work is done.

Methodology of the proposed work

The methodology of the work is detailed in Fig. 1.

Data description

Historical weather data for the study is obtained from National Solar Radiation Database (NSRDB). NSRDB is maintained by the National Renewable Energy Laboratory (NREL), by United States department of energy. This has a collection of meteorological data, covering locations all over the globe[37]. Data can be observed at 15 min, 30 min, and 60 min time intervals, in several formats. From a set of available meteorological features, desired features can be chosen to be included, based on the user requirement. The data can be accessed through multiple ways, including, direct NSRDB viewer, through an application programming interface (API) and amazon web services (AWS) cloud. For this study, irradiance data for a year, from November 2018 to October 2019, for Bangalore, Karnataka, India, observed at one hour interval, is used. The dataset includes date and time information, temperature and weather parameters. The temperature and weather parameters are tabulated in Table 1.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

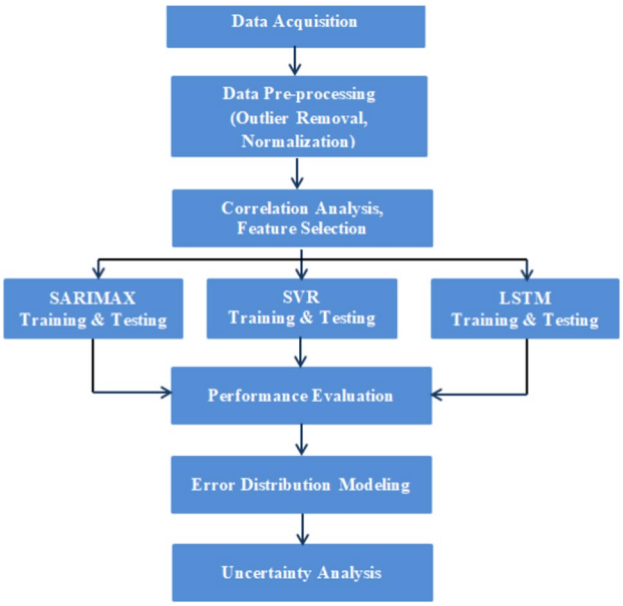


Figure 1. Methodology of the proposed work.

Parameter	Unit
Solar zenith angle	Degree
Temperature	Celsius
Pressure	mbar
Cloud type	–
Relative humidity	%
Surface albedo	–
Global horizontal irradiance (GHI)	(W/m^2)

Table 1. Parameters in the dataset.

Data preprocessing

Data analysis

Solar irradiance is highly stochastic, varying throughout the day, from early morning hours to evenings, throughout the seasons and throughout the years. The day-night cycle introduces a strong seasonal behaviour in the data. The hourly intensity is seasonal, owing to the revolution of the Earth. The daily amplitude also changes over years. All these result in a time-series with highly complex seasonality. Along with this seasonality, other factors like weather conditions, cloud cover, add additional variability in this data, thereby increasing the complexity.

The hourly variation of irradiance throughout the day is shown in Fig. 2. Similarly, Fig. 3 shows the mean irradiance for each month for the study period considered.

Solar irradiance is maximum during the month of March and lowest during the month of August. Hence, for better understanding and conclusions, the duration under study is divided into four sections, according to the seasons of the study location. Seasonal partitioning of the dataset is described in Table 2.

Mean irradiance for the four seasons is listed in Table 3. Irradiance is highest during March 2019 to May 2019, which is summer season in the area. The least is recorded during June 2019 to August 2019, which is the monsoon season. The seasonal mean irradiance is plotted in Fig. 4.

Outlier removal

Outlier removal using Inter Quartile Range (IQR) method, which uses the 'five number summary', is employed in this work[38]. This is based on the five statistical parameters in a data distribution: minimum, maximum, median, first-quartile (Q1) and third-quartile (Q3) values. IQR is the difference between third and first quartiles of the distribution. The limit value calculation is as given in Eqs. (1) and (2).

$$L_1 = Q1 - 1.5 * IQR \tag{1}$$

$$L_2 = Q3 - 1.5 * IQR \tag{2}$$

where L_1 is the lower limit and L_2 the upper limit. The data points outside these values are treated as outliers, and will be capped with the respective limit values.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

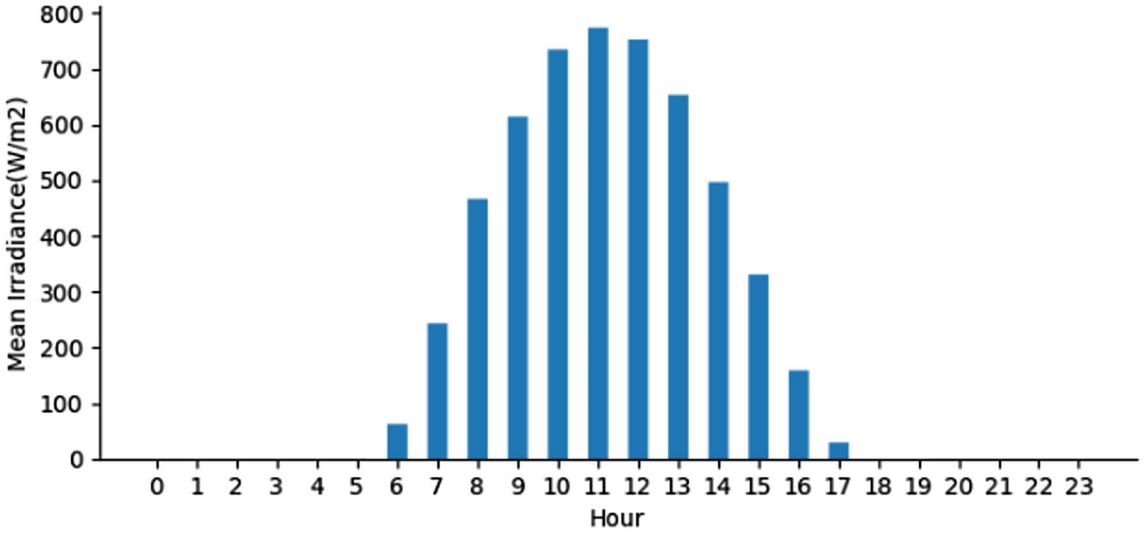


Figure 2. Hourly mean irradiance, showing the highest for hours near and at noon.

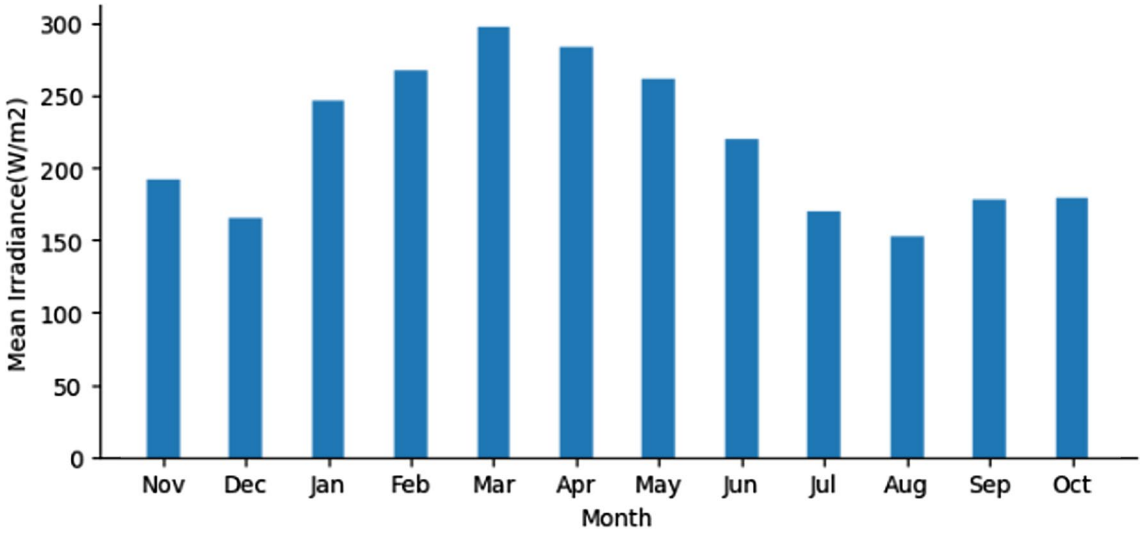


Figure 3. Monthly mean irradiance, showing the highest for March.

Dataset	Month	Seasons	Number of records
1	November2018–February2019	Winter	2880
2	March 2019–May 2019	Summer	2208
3	June 2019–August 2019	Monsoon	2208
4	September 2019–October 2019	Post-monsoon	1464

Table 2. Seasonal partitioning of the dataset.

Normalization

Among the several normalization techniques, we employed Min-Max normalization, given by Eq. (3). All the feature values are converted to a range of [0, 1].

$$x_{i,norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{3}$$

where x_i is the value of the *ith* feature before normalization and $x_{i,norm}$ is the value after normalization. x_{max} is the maximum value of the feature vector and x_{min} is the minimum value.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

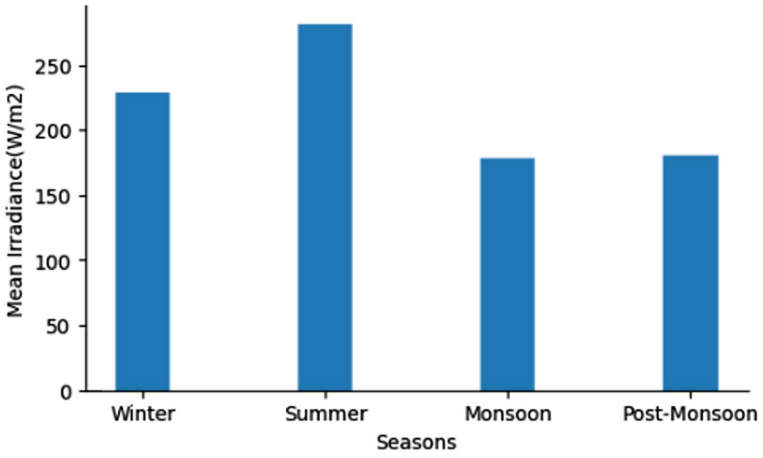


Figure 4. Seasonal mean irradiance, showing the highest for summer season.

Dataset	Season	Average irradiance (W/m^2)
1	November 2018–February 2019	228.8
2	March 2019–May 2019	281.22
3	June 2019–August 2019	178.62
4	September 2019–October 2019	180.9

Table 3. Mean seasonal irradiance.

Correlation analysis and feature selection

The interdependence between the features and the target is studied using correlation analysis. Pearson's correlation is applied and the resultant matrix is as shown in Fig. 5.

The target variable GHI is strongly influenced by the feature 'Temperature'. Also, it has strong negative correlations with 'Relative Humidity' and 'Solar Zenith Angle'. However, the features 'Surface Albedo' and 'Pressure' have very less correlation with the target. Hence these two features are removed for further analysis.

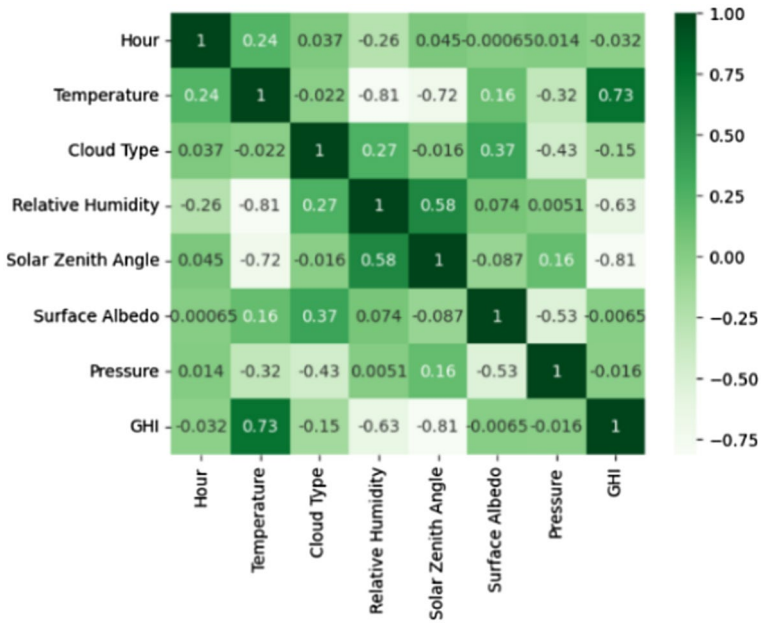


Figure 5. Confusion matrix of the dataset features.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Data partitioning
For each dataset, 80% records are employed for training and validation, and remaining 20% records for testing. This implies that, for the winter season, out of 2880 records, 576 records are used for testing. For the summer and monsoon season, out of 2208, 442 records are used for testing. For post-monsoon dataset, out of 1464 records, 288 are used for testing.

Forecast algorithms

The forecast algorithms used are detailed here:

SARIMAX
SARIMAX is an improved auto-regressive (AR), integrated, moving-average (MA) technique considering seasonal and exogenous factors. It is an improvement over ARIMA, ARIMAX and SARIMA models. SARIMAX (p, d, q) ∗ (P, D, Q, s) can be expressed as

$$\phi_p(G)\phi_p(G^s)(1-G)^d(1-G^s)^D X_t = \alpha_k y_{k,t} + \gamma_q(G)\omega_Q(G^s)\exp^t \tag{4}$$

ϕ_p and $\gamma_q(G)$ represent the non-seasonal AR and MA characteristic polynomial. $\phi_p(G^s)$, $\omega_Q(G^s)$ denotes the seasonal AR and MA polynomials. G is the backshift operator, (1-G) and $(1-G^s)$ indicate these are differencing components. The differencing degree for non-seasonal and seasonal components is d and D respectively. X_t is the value observed at instant t, $\exp^t$ is the error term and s is the seasonal pattern length. α_k is the coefficient of correlation of *kth* exogenous input variable and $y_{k,t}$ is the corresponding observed value[39,40].

SVR
SVM is a popular ML algorithm, used for regression and classification purposes. The terminology, when applied for classification tasks is support vector classifier (SVC), and for regression is, SVR[41,42]. Consider an n-dimensional dataset $\{(x_1, y(x_1))...(x_i, y(x_i)).....(x_n, y(x_n))\}$. Here, $y(x_i)$ will be 1 or -1 depending on whether x_i belongs to class 1 or class 2. The operation of SVM requires solving the optimization given in Eq. (5).

$$\min(w, b, \zeta)0.5w^T w + C\sum_{i=1}^{n}\zeta_i$$

subject to:

$$y(x_i)(w^T\phi(x_i) + b \geq 1 - \zeta_i) \tag{5}$$

where C > 0 is the penalty factor of the error, w the weight, b the bias and ζ_i the slack variable. ϕ is the function which transforms the vectors x_i to a higher dimensional space. Kernel function is given in (6).

$$K(x, x^{'}) = \phi(x)^T\phi(x)^{'} \tag{6}$$

Linear, radial basis function (RBF) and polynomial kernels are the basic kernel functions, several new kernel functions are also being proposed.

LSTM
LSTM is a modification over RNN, with memory and gate mechanisms. This is suitable when RNN fails to learn from long term dependencies[43]. Figure 6 shows the internal architecture of LSTM[44], where 'm' is the internal memory unit, 'f' the forget gate, 'o' the output gate and 'i' the input gate.

At any point of time t, working of the units are as in Eqs. (7)–(12).

$$f_t = sigm(w_f(x_t + h_{t-1}) + b) \tag{7}$$

$$i_t = sigm(w_i(x_t + h_{t-1}) + b) \tag{8}$$

$$o_t = sigm(w_o(x_t + h_{t-1}) + b) \tag{9}$$

$$m_t = sigm(w_m(x_t + h_{t-1}) + b) \tag{10}$$

$$c_t = f_t * c_{t-1} + i_t * m_t \tag{11}$$

$$h_t = o_t * \tanh(c_t) \tag{12}$$

where w_f, w_i, w_o, w_m and b are the associated weights and the bias.

Performance metrics

Numerous error metrics have been developed for forecast model evaluation. Proper selection of these metrics is important for the evaluation of model performance. A critical analysis of the error metrics used in recent studies shows that RMSE is the most popular measure for solar forecasting[45]. RMSE penalises larger errors more, which

Content courtesy of Springer Nature, terms of use apply. Rights reserved

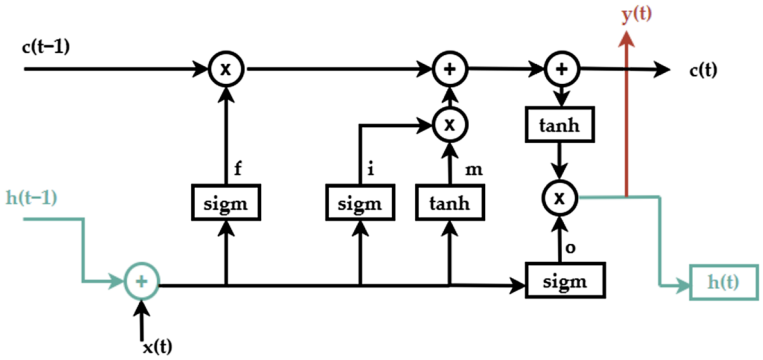


Figure 6. LSTM architecture.

are undesirable in irradiance forecasting. While dealing with different scales of data, a normalised version of RMSE, NRMSE is used. Other popular metrics are MAE and R-squared error. The metric MAPE is not suitable for solar forecasting. Solar irradiance has zero values from late evening to early morning hours. These zero values will give rise to unrealistically higher values for MAPE, which will not be a good indication of the model performance. Another useful metric is maximum error (ME), which provides information about the maximum error made by the model. In this work, forecast performance is evaluated using ME, MAE, RMSE, NRMSE and R-squared error metrics. For an n-record dataset, with y_i as the actual value of the target, $\widehat{y_i}$ as the predicted value and y_{mean}, the mean value of the target, the error metrics can be defined as below:

ME
This is the maximum error present in the model for any observation.

$$ME = max(y_i - \widehat{y_i}) \tag{13}$$

MAE
MAE is the mean magnitude of the prediction errors. MAE has the same unit as the target variable.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \widehat{y_i}| \tag{14}$$

RMSE
It is the square root of the mean of the squares of errors. It penalises larger errors more, and hence sensitive to outliers. RMSE also has the same unit as the target variable.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \widehat{y_i})^2}{n}} \tag{15}$$

NRMSE
Comparison of models using different scales of data can be made effective by normalising RMSE using mean or standard deviation of the target. In this work RMSE is normalised using the mean value.

$$NRMSE = \frac{RMSE}{y_{mean}} * 100 \tag{16}$$

R-squared
R-squared or (R^2) explains the amount of variability in the target feature, that is captured by the forecasting model. This metric is irrespective of the context or the data and hence a better measure for model comparison.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \widehat{y_i})^2}{\sum_{i=1}^{n} (y_i - y_{mean})^2} \tag{17}$$

Forecast model implementation

Methodology of the work is already detailed in Fig. 1. The prediction models are developed using Keras and Tensorflow libraries in Google Co-Lab.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

SARIMAX

SARIMAX class from 'statsmodels' package is used for building the model. A multivariate model is developed with 'Hour', 'Temperature', 'Cloud Type', 'Relative Humidity', 'Solar Zenith Angle' as the exogenous features. Manual trials were carried out to obtain the best values for 'order' and 'Seasonal order' parameters. The optimal values obtained are listed in Table 4.

SVR

SVR class from sklearn package is used for developing the model. 'Hour', 'Temperature', 'Cloud Type', 'Relative Humidity', 'Solar Zenith Angle' along with the 'GHI' values for the previous 24 hours are included as the independent features. Grid search is carried out to find the best values of hyperparameters.

LSTM

Multivariate LSTM model with 'Hour', 'Temperature', 'Cloud Type', 'Relative Humidity' and 'GHI' as features is considered. A lag of 24 is selected after multiple trials, i.e., the irradiance of any time instant is affected by the feature values of the previous 24 steps. The performance of DL model depends on the right selection of hyperparameters. In this work manual selection of parameters is carried out. Extensive trials are conducted for finalising the hyperparameters for the LSTM model. 80% of the dataset is used for training and tuning of parameters. Models with different numbers of LSTM layers and different numbers of neurons are developed and performance validated. A model with two LSTM layers, each with 50 neurons, depicted the best performance for all seasons. Table 4 gives the final model configurations selected.

Among the three models, SVR algorithm is provided here for reference. SARIMAX and LSTM also follow similar implementation.

BEGIN
Dataset : Load Solar Dataset for 2018 and 2019
Check for null values, remove if any
Group the dataset into four groups based on seasons
FOR each group
Calculate IQR & remove outliers
Remove all unwanted features
Set lag = 24
Shift the target variable (irradiance) so that value at any instant is dependent on the previous 24 values
Transform the dataset using min-max scaler
Split the dataset into train and test; (x_train, y_train) & (x_test, y_test)
Perform GridSearch to arrive at optimal values for kernel, regularization and gamma parameters
Set kernel = linear, C = 1, gamma = 1
Train the model using (x_train, y_train)
Perform model testing on (x_test, y_test)
Inverse Transform the prediction results using min-max scaler
Evaluate prediction accuracy using accuracy metrics, MAE, RMSE, NRMSE, R^2 and ME
END FOR
END

Forecast results

In all the four datasets, 80% records are used for training and validation and remaining 20% used for testing. The test results for all the four seasons are discussed in this section. The error metrics of all the models for all seasons, are tabulated in Tables 5, 6, 7 and 8. Figures 7, 8, 9 and 10 illustrates the model forecasts along with the actual values.

Model	Parameter	Value			
		Dataset1	Dataset2	Dataset3	Dataset4
SARIMAX	Order	1,0,1	1,0,1	1,1,1	1,0,1
	Seasonal order	1,1,1,24	1,1,1,24	1,1,0,24	1,0,1,24
SVR	Kernel	Linear			
	C	1			
	Gamma	1			
LSTM	LSTM layers	2			
	Neurons in each layer	50			
	Epochs	50	100	100	70
	Learning rate	0.001			
	Optimizer	Adam			
	Batch size	24			
	Output activation	Linear			

Table 4. Model parameters and values.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Winter season

Figure 7 plots the irradiance for the last week of February 2019. Actual irradiance and predictions by the three models are shown. Compared to SARIMAX and SVR models, LSTM could not predict the noon peak power for most of the days.

The error metrics for the season is tabulated in Table 5. With regard to the metrics, both MAE and RMSE values are seen better for the SVR model. For MAE, SVR shows a 5.4% reduction compared to LSTM, whereas, for RMSE, SVR shows a 10% reduction compared to SARIMAX and 12% compared to LSTM. The uncertainty involved in renewable energy sources, especially in the case of solar irradiation, is large and is very evident. On 9*th* Feb 2019 14:30 h, irradiance dropped from 461 W/m^2 to 62 W/m^2, the lowest for the hour, in the season. None of the models were able to predict this behaviour. As a result, the maximum error for all the models occurred at this hour.

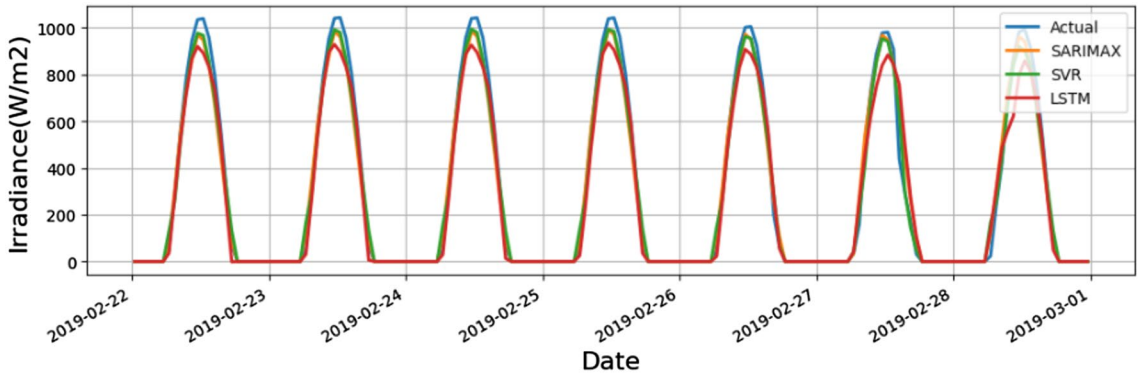


Figure 7. Actual and predicted irradiance for the last week of winter season.

Model	ME	MAE	RMSE	NRMSE	R^2
SARIMAX	529.45	32.6	65.1	24.1	0.96
SVR	458.15	33.1	58.1	21.5	0.97
LSTM	386.98	35	66	24.3	0.96

Table 5. Error metrics for winter season.

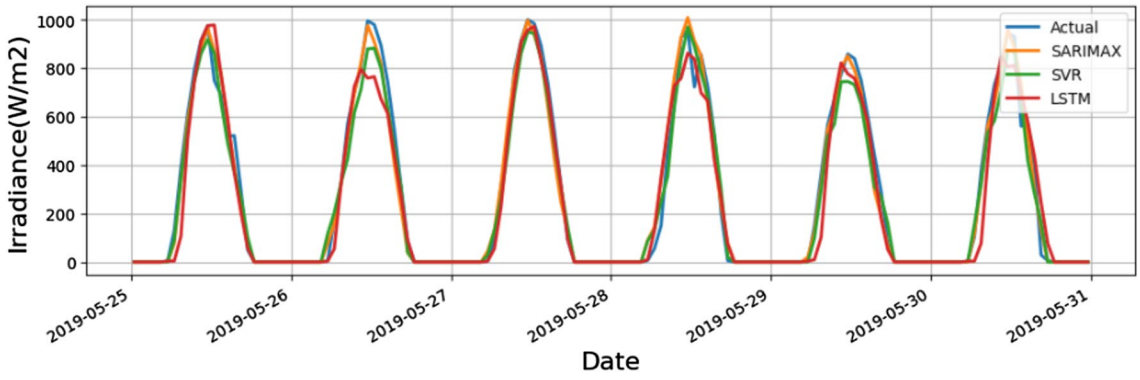


Figure 8. Actual and predicted irradiance plotted for the last week of summer season.

Model	ME	MAE	RMSE	NRMSE	R^2
SARIMAX	624.5	33	77	29.3	0.95
SVR	545	37	72	27.4	0.96
LSTM	676.5	48	103	39.2	0.91

Table 6. Error metrics for summer season.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

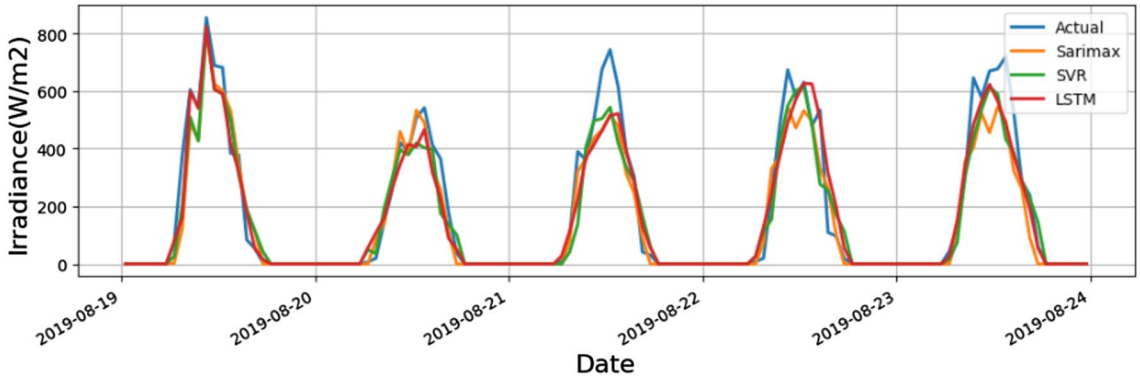


Figure 9. Actual and predicted irradiance plotted for a week of monsoon season.

Model	ME	MAE	RMSE	NRMSE	R^2
SARIMAX	413	51	98	60	0.83
SVR	501.89	48	92	56.3	0.85
LSTM	518	60	119	72.8	0.75

Table 7. Error metrics for monsoon season.

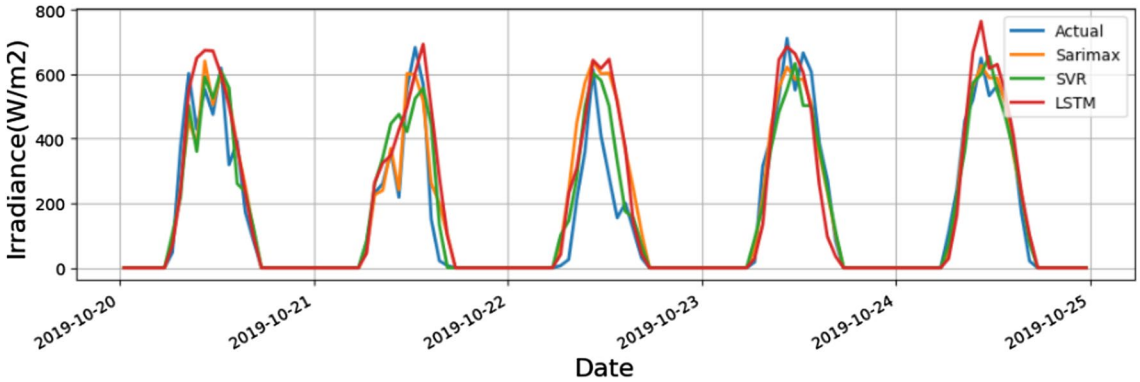


Figure 10. Actual and predicted irradiance plotted for a week of post-monsoon season.

Model	ME	MAE	RMSE	NRMSE	R^2
SARIMAX	440	46	92.3	62.1	0.82
SVR	366.6	45.6	85.6	57.6	0.85
LSTM	608	54	108	72.7	0.76

Table 8. Error metrics for post-monsoon season.

Summer season

Figure 8 plots the irradiance for the last week of May 2019. Even though all the models track the trends in the data, prediction errors are more compared to winter season.

From the MAE perspective, SARIMAX shows a reduction of approximate 10% compared to SVR and 31% compared to LSTM model. However, from the RMSE perspective, SVR shows better performance with 6.5% reduction compared to SARIMAX and 30% reduction compared to LSTM.

Monsoon season

Figure 9 shows the prediction results for monsoon season, plotted for the last week of August. Prediction errors are more significant in this season compared to winter and summer seasons. From MAE and RMSE perspective, SVR shows significant improvement compared to SARIMAX and LSTM models.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Post-monsoon season

Figure 10 shows the prediction results for post-monsoon season, plotted for the third week of October. Similar to the monsoon period, prediction errors are more compared to winter and summer seasons. On the error metrics perspective, SVR shows better performance compared to SARIMAX and LSTM models.

Figures 11 and 12 give a graphical comparison of MAE and RMSE metrics. For all the seasons, MAE and RMSE values are the highest for LSTM model. In terms of MAE, SARIMAX and SVR showed comparable performances. For RMSE, however, the SVR model showed the least errors, for all the seasons.

Uncertainty analysis

The above-discussed models, SARIMAX, SVR and LSTM techniques predict one single value for the future, i.e. irradiance value for a particular hour. However better the point forecasting models perform, owing to the high variability in the nature of solar irradiance, there will be a factor of uncertainty in the forecasts. From a system operation perspective, in order to obtain accurate forecasts, the uncertainty also should be taken into account. Therefore, providing a range or interval, where the value will possibly lie, will be more suitable than predicting a single value. A comparison of the proposed work with two of the popular uncertainty related literature is tabulated in Table 9.

In this work, prediction uncertainty is quantified using CIs and coverage rates. The forecast error histogram is analysed to obtain the nature of the distribution. Further, confidence limits of the distribution for multiple levels and the associated coverage rates, give information about the uncertainty involved in the model forecast. The model performance with coverage rates greater than the confidence level is preferred.

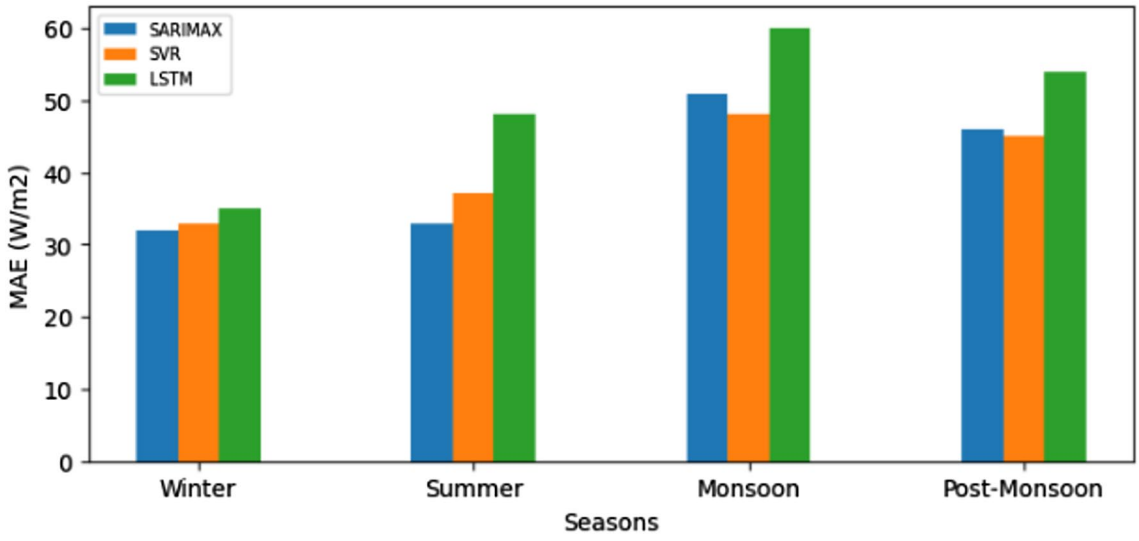


Figure 11. MAE metrics for all the models.

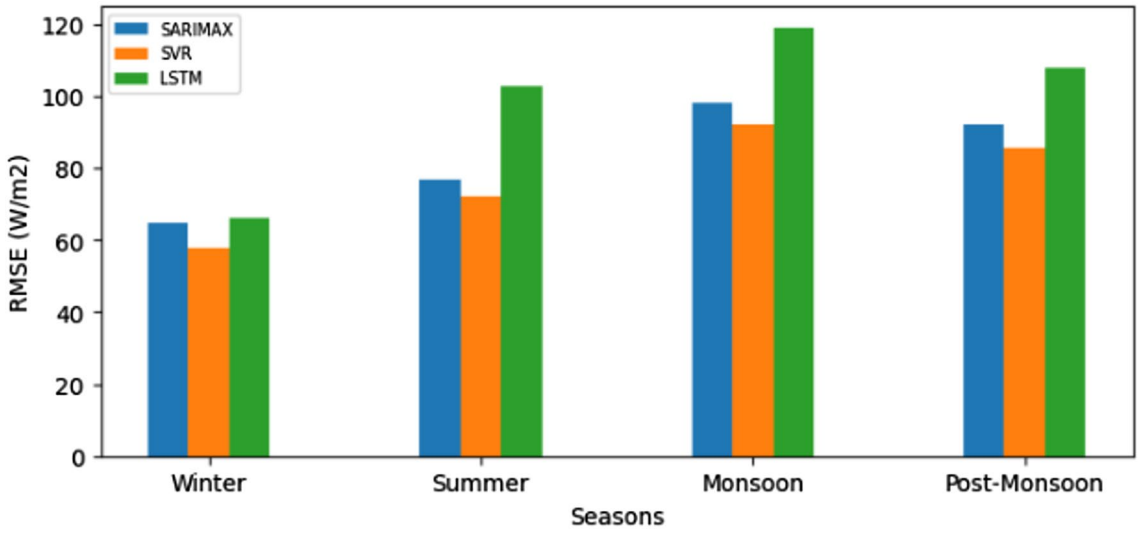


Figure 12. RMSE metrics for all the models.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Parameter	Ref.[46]	Ref.[47]	Proposed work
Predictor variable	Solar irradiance	Solar power	Solar irradiance
PointForecasting technique	SVR	CNN-BiLSTM	SARIMAX, SVR, LSTM
IntervalForecasting technique	Predicted and measured irradiance values are converted to clearness index. Multivariate normality assumption of the joint distribution of clearness indices is made	Statistical distribution of the model forecast errors is analysed	Statistical distribution of SVR forecast errors is analysed
Error probability distribution	–	Gaussian mixture model	Laplace
Uncertainty estimation	Confidence intervals, hit rate	Confidence intervals, coverage rate	Confidence intervals, coverage rate

Table 9. Comparison with two popular works on prediction uncertainty.

Forecast error distribution fitting

The probability density function describes a range of values for a random variable and the chances that a sample falls in a given interval. This concept is used to quantify uncertainty in this work. For this, a model which can represent the error distributions accurately, needs to be found out. The error histograms for SVR forecast for all the seasons are plotted as shown in Fig. 13 and the probability density distribution is analysed. All the distributions are peaked, with narrower shoulders and longer tails, indicating a not-normal behaviour. However, this has to be verified by statistical means.

From the initial visual inspection of the distribution and the literature studies conducted, normal distribution, gamma distribution, weibull distributon, symmetric and asymmetric laplace distributions are selected for our analysis. The parameters and range for each of these distributions are tabulated in Table 10[48].

All the selected distributions are fit to the error data by maximum likelihood estimation. The distribution parameters obtained are detailed in Table 11.

The best-fit distribution is found considering several factors, which also depend on the context of the application. In previous literature, statistical tests which include Anderson–Darling test, Kolmogorov–Smirnov (KS) test, and quantile–quantile (Q–Q) plots, are used to verify the correctness of a fit[30].

In this work, the goodness of fit is evaluated using sum-square error and KS test. The error data is fitted to each of the distributions and the sum of squares of error in the fit is calculated. The distribution with the least

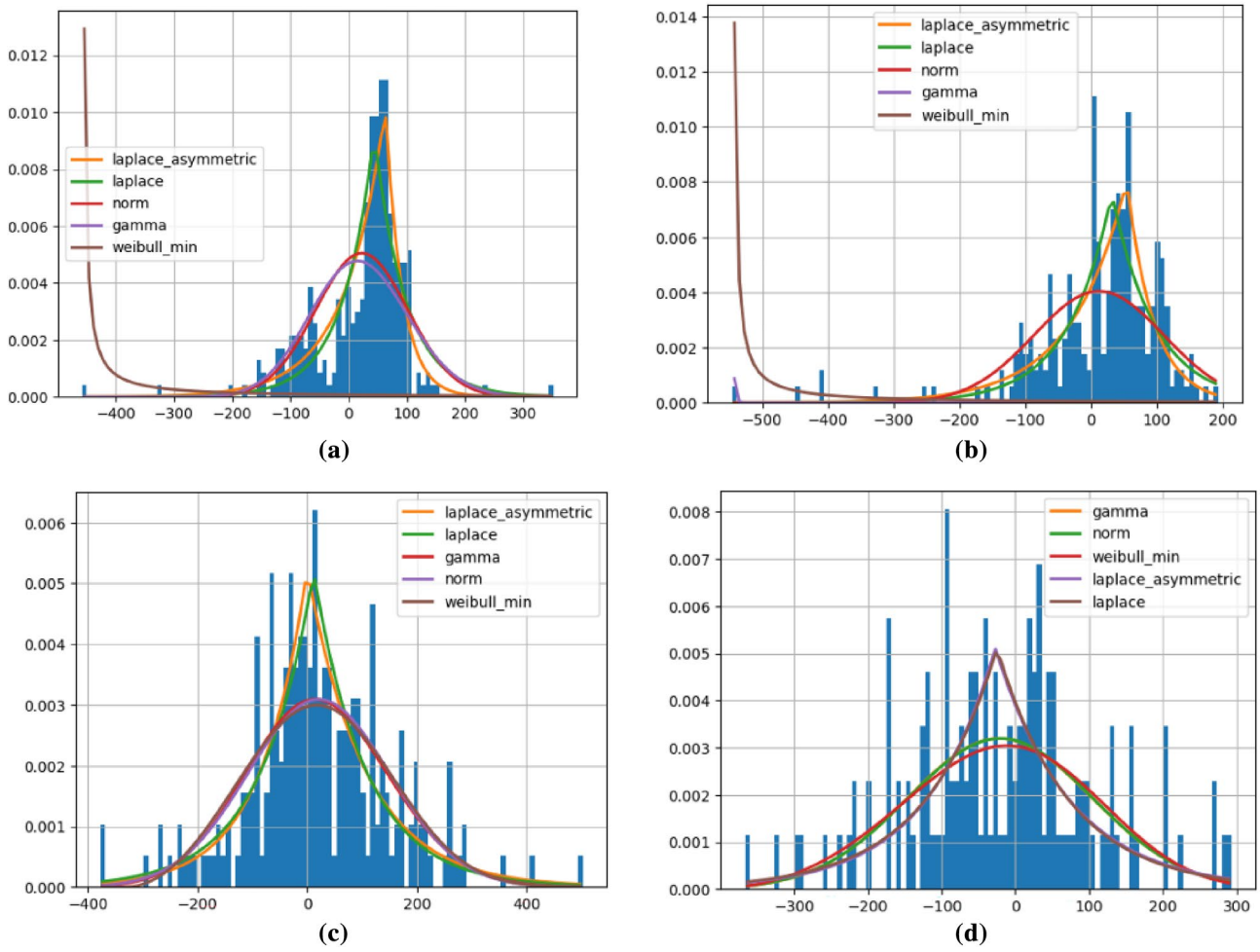


Figure 13. Error distribution for all the seasons. (**a**) Winter, (**b**) summer, (**c**) monsoon, (**d**) post-monsoon.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

error can be considered as the best fit. In this manner, laplace distribution is found to be suitable for the error plots. The distribution for winter is a skewed one; hence it is fitted using asymmetric laplace distribution. Asymmetric laplace distribution is similar to the laplace, except with a shape parameter 'kappa' which, in this case, is 1.549. For summer, monsoon and post-monsoon, the error distributions are found to match symmetric laplace distribution. This is further verified using the KS test. KS test is a non-parametric method used to determine the similarity of two distributions, or if a given distribution is similar to a hypothesized distribution[49]. This test statistic measures the vertical distance between the empirical cumulative distribution function of the data and the cumulative distribution function of the theoretical distribution being tested. For any data belonging to the specified theoretical distribution, test statistics will be small. KS test statistics along with the p-values for all seasons are tabulated in Table 12. These results confirm the suitability of laplace distribution for fitting the error data.

Confidence band construction

Confidence bands for multiple confidence levels, 95%, 90%, 85% and 80%, for laplace distribution, are constructed for different seasons. For an irradiance forecast value x, an interval, $x_1^{\beta} < x < x_2^{\beta}$, for a given probability (1-β), $0 < \beta < 1$, $P(x_1^{\beta} < x < x_2^{\beta}) = 1\text{-}\beta$, the interval is called a (1-β)100% CI. 1- β is the degree of confidence, and the end points are called the lower and upper confidence limits[50]. Confidence limits for the selected confidence levels for all seasons are calculated and tabulated in Tables 13 and 14.

Figures 14, 15, 16 and 17 plots the confidence bands for different seasons. The forecasted and actual values are plotted in blue and black colours respectively. The region between the two red lines represents the 80% confidence band. Outside, on both sides of the red lines, the two yellow lines represent the 85% confidence level. The two cyan lines represent 90% confidence and the outermost pink lines represent the 95% confidence band. It can be seen that as the confidence level increases, bandwidth also increases. 95% has a broader confidence band than the 80% level. A narrow confidence band is always desirable for a good forecast.

The actual and the predicted values, along with confidence bands, for 28*th* Feb 2019 are plotted in Fig. 14. Similar to the discussion in the previous section, the actual and predicted values are close. Most of the actual values fall in the 95% band which is the broadest, whereas, the 80% band has the least number of actual values.

Distribution	Definition	Parameters	Range
Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\mu - Mean, \sigma - StandardDeviation$	$(-\infty, \infty)$
Weibull	$\frac{\gamma}{\alpha}\left(\frac{x-\mu}{\alpha}\right)^{\gamma-1} e^{-\left(\frac{x-\mu}{\alpha}\right)^{\gamma}}$	$\gamma - Shape, \alpha - Scale, \mu - Location$	$[0, \infty)$
Gamma	$\frac{x^{\alpha-1}}{\Gamma(\alpha)\theta^{\alpha}} e^{-\left(\frac{x}{\theta}\right)}$	$\alpha - Shape, \theta - Scale$	$[0, \infty)$
Laplace	$\frac{1}{2b} e^{-\frac{\lvert x-\mu\rvert}{b}}$	$\mu - location, b - Scale$	$(-\infty, \infty)$

Table 10. Parametric estimation specifications.

Distribution		Winter	Summer	Monsoon	Post-monsoon
Normal	Mean	66.29	70.88	96.7	98.65
	Std. deviation	48.57	69.3	87.87	79.08
Weibull	Shape	1.46554	1.14	1.03	1.2
	Scale	73.19111	74.63	98.04	104.7
	Loc	0	0	0	0
Laplace	Scale	44.85	65.94	96.19	97.796
	Loc	63.1781	31.54	11.393	-24.98
Gamma	Scale	20.15	52.19	65.97	72.57
	Loc	58.79195	0	0	0

Table 11. Distribution parameters for all seasons.

Seasons	Test statistics	Test p-value
Winter	0.071	0.103
Summer	0.082	0.07
Monsoon	0.056	0.466
Post-monsoon	0.07	0.39

Table 12. KS test statistics.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Seasons	Confidence level: 95%		Confidence level: 90%	
	Lower	Upper	Lower	Upper
Winter	−71.33	197.69	−51	177.63
Summer	−197.5	260.64	−151	214
Monsoon	−288	310	−221	244
Post-monsoon	−292	242	−225	175

Table 13. Confidence interval limits for 95% and 90% confidence levels.

Seasons	Confidence level: 85%		Confidence level: 80%	
	Lower	Upper	Lower	Upper
Winter	−39.53	165.9	−31.21	157.56
Summer	−125	188	−106	169
Monsoon	−182	205	−154	177
Post-monsoon	−185	135	−157	107

Table 14. Confidence interval limits for 85% and 80% confidence levels.

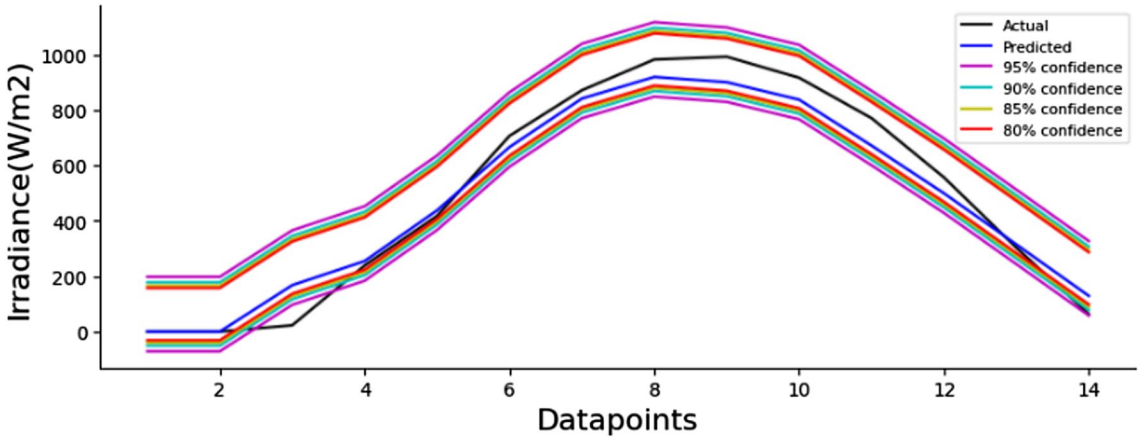


Figure 14. Confidence bands for a day in winter for different confidence levels.

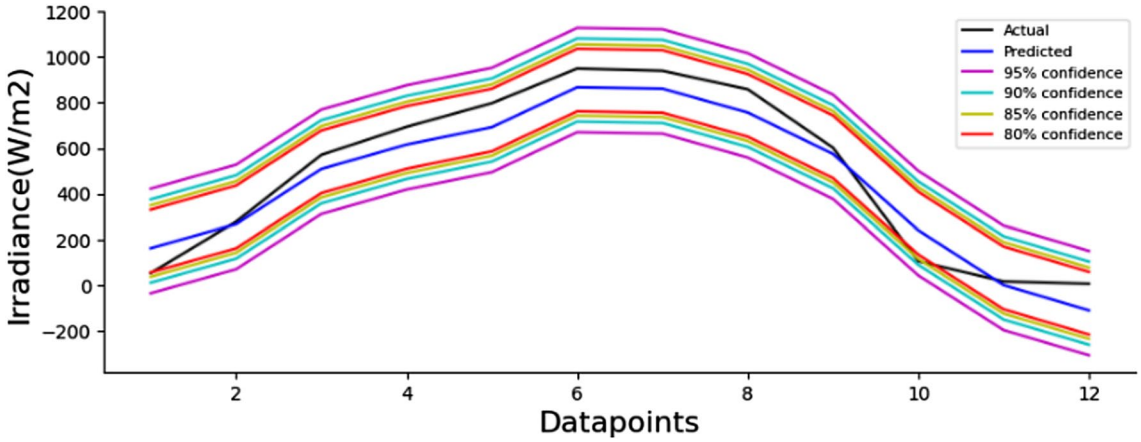


Figure 15. Confidence bands for a day in summer, for different confidence levels.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

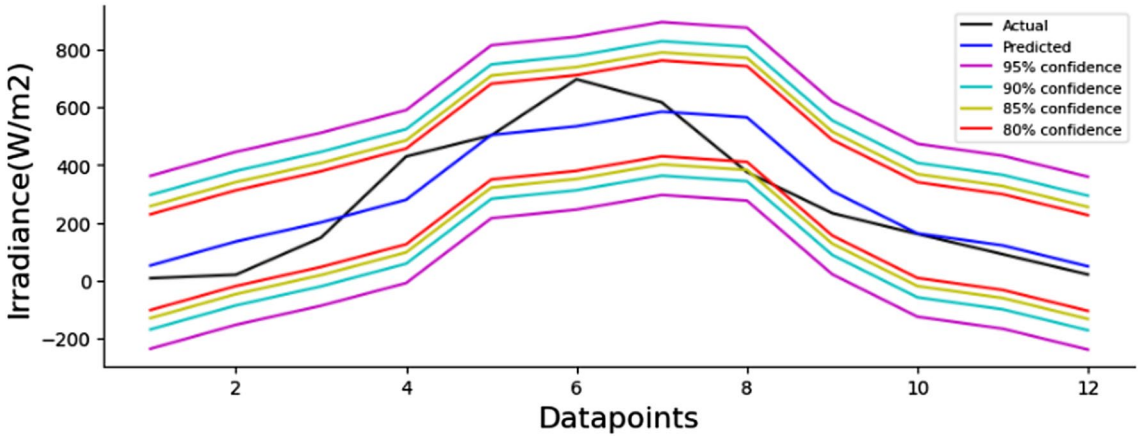


Figure 16. Confidence bands for a day in monsoon, for different confidence levels.

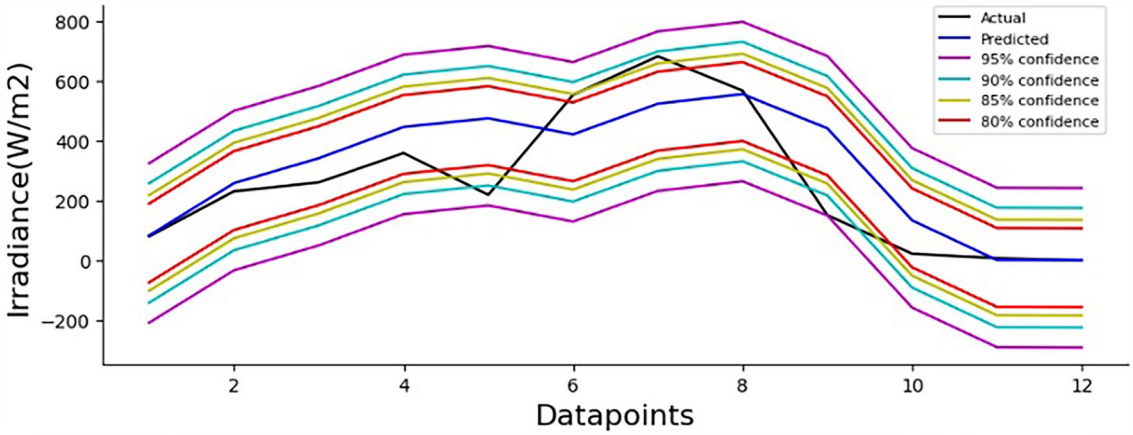


Figure 17. Confidence bands for a day in post-monsoon, for different confidence levels.

Figures 15 and 16 show the confidence bands for selected days in summer and monsoon seasons, plotted with actual and forecasted values. In both the cases, most of the actual values fall within the 80% confidence band, which shows a good distribution modelling.

For a selected day in the post-monsoon season, as shown in Fig. 17, most of the actual values are outside the 80% band and 85%, but still inside the 95% confidence level.

Forecast coverage rate

Coverage rate is the number of actual values within a certain confidence band, expressed as a percentage of the total observations[51], as given in Eq. (18). Higher the coverage rate, better the error distribution modeling. Coverage rates for the complete test period, for all the confidence levels are tabulated in Table 15.

$$Coverage\ Rate = \frac{No.\ of\ values\ within\ the\ band}{Total\ Number\ of\ values} \tag{18}$$

As seen from Table 15, as the confidence level decreases, coverage rate also decreases. This is because the band becomes narrow and, more and more actual values fall out of the band. Statistically, a narrow confidence band gives the best prediction. However, narrow confidence bands implicate small coverage rates. Hence, a compromise is required between the choice of confidence level and the coverage rates.

For all the cases listed in Table 15, except the first, coverage rates are well above the corresponding confidence levels. This shows that the selected laplace distribution accurately models the errors and can provide a reliable measure of the uncertainty associated with the forecasts.

Conclusion and future scope

The work combines point forecasting and interval forecasting, with a view to quantify the solar forecasting uncertainty. Forecast errors are analysed and characterised using suitable statistical distribution. Further, the uncertainty is quantified by means of CIs and coverage rates. In the first stage, the work forecasts the solar irradiance at Bangalore, Karnataka, India using three methods. Considering the seasonal influence on irradiance, the data under study is divided into four seasons, summer, winter, monsoon and post-monsoon, in accordance with

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Seasons	Coverage rate			
	CI: 95%	CI: 90%	CI: 85%	CI: 80%
Winter	93	90.3	89.7	89
Summer	98.4	97.9	97	95
Monsoon	98.6	95	93.4	90.9
Post-monsoon	97.6	95	91.6	88

Table 15. Coverage rates for different seasons for different confidence levels.

the location under study. Time Series, ML and DL techniques are employed and forecasting is done for different seasons. Model performance is evaluated using error metrics ME, MAE, RMSE, NRMSE and (R^2). Among the models, SVR model is found to exhibit the best performance for all the seasons. Following the forecasting stage, uncertainty analysis is carried out. The forecast error data is found to follow laplace distribution in all seasons. Confidence bands for multiple CIs are constructed and coverage rates are calculated and presented. For all the scenarios, excellent coverage rates are obtained for the considered confidence levels. This confirms the right selection and fitting of the error distribution.

An efficient renewable integrated microgrid or distributed generation system requires accurate planning and control of resources. A reliable supply and demand forecasting system is the primary step towards this direction. Over the years, a large variety of approaches have proliferated to provide point forecasts of renewable energy. However, most of the forecasting works have not analysed the uncertainty associated with these forecasts. This work analyses the aforementioned uncertainty through CIs and coverage rates computed by means of parametric approach. This integrated approach of uncertainty estimation can lead to robust system performance.

As a future work, different confidence levels and associated coverage rates can be explored. A model with narrow confidence band and coverage rate close to the confidence level gives the best forecast. At the next level, these findings can be used for producing accurate interval forecasts for a variety of applications. In the context of our work, these findings can be used for quantifying uncertainty as a percentage and used for the power flow optimization in a renewable integrated microgrid.

Data availability

The dataset analysed for this study is available in the NSRDB repository, https://nsrdb.nrel.gov/data-viewer. Providing the location address, several available datasets can be obtained. The dataset 'Europe, Africa & Asia (15, 30, 60 min / 4 km / 2017–2019)' is used for the study.

Received: 23 May 2024; Accepted: 24 July 2024
Published online: 02 August 2024

References

1. Kolkowska, N. *Grid Integration of Renewable Energy. Sustainable Review*. https://sustainablereview.com/grid-integration-of-renewable-energy.
2. Thejus, S. & P, S., Deep learning-based power consumption and generation forecasting for demand side management. In *2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India*. 1350–1357 (2021).
3. Vinothine, S., Arachchige. L. N. W, Rajapakse, A. D. & Kaluthanthrige, R. Microgrid energy management and methods for managing forecast uncertainties. *Energies* **15**, 8525 (2022).
4. Boralessa, M. A. K. S., Hovden, S., Wickramarathna, A. V. U. A., & Hemapala, K. T. M. U. Effect of renewable energy forecasting error on model predictive control based microgrid energy management system. In *2022 IEEE IAS Global Conference on Emerging Technologies* (*GlobConET*) (2022).
5. Akhter, M. N., Mekhilef, S., Mokhlis, H. & Shah, N. M. Review on forecasting of photovoltaic power generation based on machine learning and metaheuristic techniques. *IET Renew. Power Gener.* **13**, 1009–1023 (2019).
6. Wang, G., Su, Y. & Shu, L. One-day-ahead daily power forecasting of photovoltaic systems based on partial functional linear regression models. *Renew. Energy* **96**, 469–478 (2016).
7. Singh, B., Pozo, D. A. Guide to solar power forecasting using ARMA models. In *IEEE PES Innovative Smart Grid Technologies Europe* (*ISGT-Europe*). *Bucharest, Romania*. Vol. 2019. 1–4 (2019).
8. Vishnu, G. *et al.* Short-term forecasting of electric vehicle load using time series, machine learning, and deep learning techniques. *World Electr. Veh. J.* **14**, 266 (2023).
9. Sivakumar, S. *et al.* Machine learning approach on time series for PV-solar energy. *Adv. Mater. Sci. Eng.* **2022**, 1–7 (2022).
10. Chandran, L. R., Jayagopal, N., Lal, L. S., Narayanan, C., Deepak, S. & Harikrishnan. Residential load time series forecasting using ANN and classical methods. In *2021 6th International Conference on Communication and Electronics Systems* (*ICCES*), *Coimbatore, India*. 1508–1515 (2021).
11. Ranga, J. *et al. Smart Technologies, Communication and Robotics* (*STCR*). *Sathyamangalam, India*. Vol. 2022. 1–5 (2022).
12. Preda, S., Oprea, S., Bâra, A. & Belciu (Velicanu), A. PV forecasting using support vector machine learning in a big data analytics context. *Symmetry* **10**, 748 (2018).
13. Yen, C., Hsieh, Y., Su, K., Yu, M. & Leu, J. Solar power prediction via support vector machine and random forest. *E3S Web Conf.* **69**, 01004 (2018)
14. Lee, D. & Kim, K. Recurrent neural network-based hourly prediction of photovoltaic power output using meteorological information. *Energies* **12**, 215 (2019).
15. Hossain, M. S. & Mahmood, H. Short-term photovoltaic power forecasting using an LSTM neural network and synthetic weather forecast. *IEEE Access* **8**, 172524–172533 (2020).
16. Huynh, A. N. *et al.* Near real-time global solar radiation forecasting at multiple time-step horizons using the long short-term memory network. *Energies* **13**, 3517 (2020).

Content courtesy of Springer Nature, terms of use apply. Rights reserved

17. Brahma, B. & Wadhvani, R. Solar irradiance forecasting based on deep learning methodologies and multi-site data. *Symmetry* **12**, 1830 (2020).
18. Li, G. *et al.* Photovoltaic power forecasting with a hybrid deep learning approach. *IEEE Access* **8**, 175871–175880 (2020).
19. Eom, H., Son, Y. & Choi, S. Feature-selective ensemble learning-based long-term regional PV generation forecasting. *IEEE Access* **8**, 54620–54630 (2020).
20. Ferkous, K., Guermoui, M., Menakh, S., Bellaour, A., Boulmaiz, T. A novel learning approach for short-term photovoltaic power forecasting—A review and case studies.*Eng. Appl. Artif. Intell.* **133**, Part E (2024).
21. Singla, P., Duhan, M. & Saroha, S. An integrated framework of robust local mean decomposition and bidirectional long short-term memory to forecast solar irradiance.*Int. J. Green Energy* **20**, 1073–1085 (2022).
22. Singla, P., Duhan, M. & Saroha, S. A point and interval forecasting of solar irradiance using different decomposition based hybrid models.*Earth Sci. Inform.* **16**, 2223–2240 (2023).
23. Pan, M. *et al.* Photovoltaic power forecasting based on a support vector machine with improved ant colony optimization. *J. Clean. Prod.* **277**, 123948 (2020).
24. VanDeventer, W. *et al.* Short-term PV power forecasting using hybrid GASVM technique. *Renew. Energy* **140**, 367–379 (2019).
25. Liu, Z., Li, L., Tseng, M. & Lim, M. K. Prediction short-term photovoltaic power using improved chicken swarm optimizer—Extreme learning machine model. *J. Clean. Prod.* **248**, 119272 (2020).
26. Michael, N. E., Hasan, S., Al-Durra, A. & Mishra, M. Short-term solar irradiance forecasting based on a novel Bayesian optimized deep long short-term memory neural network. *Appl. Energy* **324**, 119727 (2022).
27. Syama, S. & Ramprabhakar, J. Multistep ahead solar irradiance and wind speed forecasting using Bayesian optimized long short term memory. In *2022 7th International Conference on Communication and Electronics Systems* (*ICCES*) (2022).
28. Dawan, P. *et al.* Comparison of power output forecasting on the photovoltaic system using adaptive neuro-fuzzy inference systems and particle swarm optimization-artificial neural network model. *Energies* **13**, 351 (2020).
29. Al-lahham, A., Theeb, O., Elalem, K., Alshawi, T. A. & Alshebeili, S. A. Sky imager-based forecast of solar irradiance using machine learning. *Electronics* **9**, 1700 (2020).
30. Hodge, B., Milligan, M. Wind & power forecasting error distributions over multiple timescales. In *IEEE Power and Energy Society General Meeting. Detroit, MI, USA*. Vol. 2011. 1–8 (2011).
31. Mararakanye, N., Dalton, A. & Bekker, B. Characterizing wind power forecast error using extreme value theory and copulas. *IEEE Access* **10**, 58547–58557 (2022).
32. Oosthuizen, C. *et al. International SAUPEC/RobMech/PRASA Conference. Cape Town, South Africa*. Vol. 2020. 1–6 (2020).
33. Liu, L. *et al.* Prediction of short-term PV power output and uncertainty analysis. *Appl. Energy* **228**, 700–711 (2018).
34. Liu, Y. & Liu, L. Wind power prediction based on LSTM-CNN optimization. *Sci. J. Intell. Syst. Res.* **3** (2021).
35. Huang, Y., Li, P. & Sun, Y. Uncertainty analysis of wind power based on operating data. In *2021 4th International Conference on Energy, Electrical and Power Engineering* (*CEEPE*), *Chongqing, China*. 500–506 (2021).
36. Cao, W., Zhou, J., Xu, Q., Zhen, J. & Huang, X. Short-term forecasting and uncertainty analysis of photovoltaic power based on the FCM-WOA-BILSTM model. *Front. Energy Res.* **10** (2022).
37. NSRDB. https://nsrdb.nrel.gov/about/what-is-the-nsrdb.
38. Mahajan, M., Kumar, S., Pant, B. & Tiwari, U. K. Incremental outlier detection in air quality data using statistical methods. In *2020 International Conference on Data Analytics for Business and Industry: Way Towards a Sustainable Economy* (*ICDABI*), *Sakheer, Bahrain*. 1–5 https://doi.org/10.1109/ICDABI51230.2020.9325683 (2020).
39. Manigandan, P. *et al.* Forecasting natural gas production and consumption in United States—Evidence from SARIMA and SARIMAX models. *Energies* **14**, 6021 (2021).
40. Au, J., Saldaña, J. Jr., Spanswick, B. & Santerre, J. Forecasting power consumption in Pennsylvania during the COVID-19 pandemic: A SARIMAX model with external COVID-19 and unemployment variables. *SMU Data Sci. Rev.* **3**, 6 (2020).
41. Boser, B. E., Guyon, I. M. & Vapnik, V. N. A training algorithm for optimal margin classifiers. In *Proceedings of the Fifth Annual Workshop on Computational Learning Theory* (*COLT* '92), *Pennsylvania, USA*. 144–152 (1992).
42. Hsu, C., Chang, C. & Lin, C. *A Practical Guide to Support Vector Classification* (2008).
43. Hochreiter, S. The vanishing gradient problem during learning recurrent neural nets and problem solutions. *Int. J. Uncertain. Fuzzin. Knowl.-Based Syst.* **6**, 107–116 (1998).
44. Kriekinge, G. V., Cauwer, C. D., Sapountzoglou, N., Coosemans, T. & Messagie, M. Day-ahead forecast of electric vehicle charging demand with deep neural networks. *World Electr. Veh. J.* **12**, 178 (2021).
45. Singla, P., Duhan, M. & Saroha, S. Review of different error metrics: A case of solar forecasting. *AJSE* **20**, 158–165 (2021).
46. Murata, A., Ohtake, H., Oozeki, T. Modeling of uncertainty of solar irradiance forecasts on numerical weather predictions with the estimation of multiple confidence intervals. Renewable Energy 117, pp. 193 - 201 (2018)
47. Gu, B., Li, X., Xu, F., Yang, X., Wang, F., Wang, P. Forecasting and Uncertainty Analysis of Day-Ahead Photovoltaic Power Based on WT-CNN-BiLSTM-AM-GMM. Sustainability 15 (2023)
48. Weisstein, E. W. Laplace Distribution. https://mathworld.wolfram.com/LaplaceDistribution.html.
49. Dodge, Y. *Kolmogorov–Smirnov Test. The Concise Encyclopedia of Statistics*. 283–287 (Springer, 2008).
50. Charytoniuk, W. & Niebrzydowski, J. Confidence interval construction for load forecast. *Electr. Power Syst. Res.* **48**, 97–103 (1998).
51. Li, G., Wu, D. K., Zhou, M. & Liu, A. The combination of interval forecasts in tourism. *Ann. Tour. Res.* **75**, 363–378 (2019).

Acknowledgements

Authors express their gratitude to Amrita Vishwa Vidyapeetham for the help and support received for this work.

Author contributions

Conceptualization, G.V. and D.K.; methodology, G.V., D.K., S.R.S.; software, G.V.; validation, G.V.; formal analysis, G.V., D.K.; writing-original draft preparation, G.V.; writing-review and editing, G.V., D.K., S.R.S.; supervision, D.K.; project administration, D.K., S.R.S. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to D.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2024

Content courtesy of Springer Nature, terms of use apply. Rights reserved

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).
Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.
These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.
We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.
While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval , sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.
These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.
To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.
Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.
If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com